
Assignment 1 - Swedish Pre-A1 Text Classifier

Embedding-Based Text Classification for Pre-A1 Swedish Sentences

Project Links

GitHub: <https://github.com/SeanSha/swedish-scenario-classifier>

Dataset: <https://huggingface.co/datasets/SeanSha30/swedish-pre-a1-scenario-classifier-dataset>

Demo: <https://huggingface.co/spaces/SeanSha30/swedish-scenario-classifier-demo>

1 Introduction

The first aim of this project is to build an embedding-based text classification system for absolute beginner / pre-A1 Swedish learner texts. The second aim is to explore how Codex can be used to support the development of a small study project. Therefore, the project has two directions. The first is to classify short Swedish learner sentences into one of six lecture-aligned daily-life scenario labels: `food_shop`, `family_school`, `health_places`, `transport`, `home_places`, and `social_intro`. The second is to reflect on how Codex can help with task planning, dataset creation, coding, and documentation.

The broader goal is to create a small but useful domain dataset for Swedish language learning. In a later tutoring agent, such a dataset could help the system detect the learner's situation and select suitable practice material. If this approach works, the dataset can be further expanded for Swedish learners at different levels.

2 Dataset Creation

The custom dataset contains 150 examples, balanced across six labels with 25 examples per class. Each row includes a Swedish sentence, English translation, Chinese translation, label, lecture theme, source, and review status. The model uses only the Swedish text field as input; the translation and lecture-theme fields are metadata for review and explanation.

The dataset was created through AI-assisted drafting and manual structuring adjustment. Its vocabulary and themes were aligned with beginner Swedish lecture materials, including introductions, clarification phrases, food and shops, family and school words, health-related places, transport, and home/location prepositions. To avoid publishing course materials, the public dataset contains newly written example sentences rather than copied lecture text.

3 Method

Sentence embeddings were generated with `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2` [1], a lightweight multilingual sentence embedding model suitable for Swedish learner sentences. Four classifiers were trained and compared on the same embeddings: Logistic Regression, Linear SVM, Random Forest, and kNN. Evaluation used a stratified 80/20 train/test split with 120 training examples and 30 test examples. The main metrics were accuracy, macro-F1, per-label classification reports, and confusion matrices.

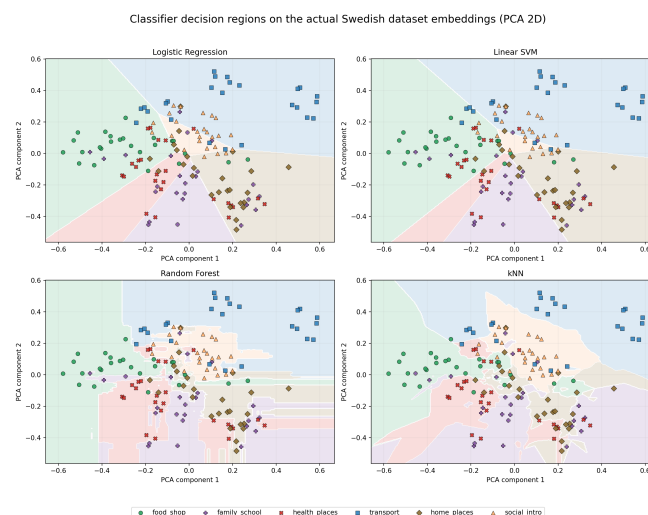


Figure 1: Simplified illustration of the classifier types used in the project. The real assignment uses high-dimensional sentence embeddings, but the figure shows the intuition: kNN classifies by neighbor voting, Logistic Regression and Linear SVM learn linear decision regions, and Random Forest creates block-like regions through tree voting.

4 Results

Classifier	Accuracy	Macro-F1
Linear SVM	0.8667	0.8712
Logistic Regression	0.8667	0.8689
Random Forest	0.8667	0.8682
kNN	0.8667	0.8648

Table 1: Classifier comparison on the test set.

All four classifiers reached the same accuracy because the test set is small: each model correctly classified 26 out of 30 examples. Macro-F1 still differs slightly because the models made different errors across labels. The strongest model by macro-F1 was Linear SVM. Error analysis showed that very short beginner sentences can be ambiguous. For example, *Jag bor i Stockholm* can be interpreted as either a home/place sentence or a social introduction, and *Är det långt?* can be a transport question but lacks explicit transport vocabulary. This suggests that future dataset expansion should add more minimal context while keeping the language pre-A1.

5 Demo and Reflection on AI Use

The online demo is hosted on Hugging Face Spaces and uses a simple Gradio web interface. At startup, it loads the dataset, generates multilingual sentence embeddings, trains a Linear SVM classifier, and predicts a scenario label for user input. This makes the system testable as a working prototype rather than only an offline experiment.

Before the coding started, the assignment context was discussed with ChatGPT and developed through several iterative discussions. The task requirements were carefully reviewed before implementation. Codex was useful for quickly drafting beginner-level examples, restructuring labels, writing reproducible experiment code, and preparing documentation.

The project contains three main source-code files. The main experiment is implemented in `run_experiment.py`, which loads the dataset, generates embeddings, trains the four classifiers, evaluates them, and saves the results. The other two scripts are `train_tool_model.py`, which trains and saves a lightweight classifier, and `predict_scenario.py`, which predicts a scenario label from a sentence. An additional file is used to build the Hugging Face demo page.

Codex was useful for quickly drafting the project idea, implementation plan, and code structure. However, all LLM-assisted work still needed careful human review. In this project, I checked the task requirements, aligned the dataset with the lecture content, added a two-dimensional classifier comparison figure, changed the system from one classifier to four classifiers, tested the Hugging Face Space demo, and reviewed the GitHub codebase. In Assignment 2, I plan to test this dataset further and explore how it can be used in a Swedish tutor agent.

References

- [1] Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics. <http://arxiv.org/abs/1908.10084>